

Published baselines (intensity only)

Covariate-using models (this work)

Event times
only

Univariate Hawkes (MLE)

LightGBM (tuned)

Market-augmented Hawkes

same TPP family, + covariates: 0.13→0.26

CovTPP (GRU hazard)

17 crowding
+ cross-asset
covariates

Transformer Hawkes (THP)

ST-GNN (cross-asset graph)

Identical
test bins
PR-AUC ↑

